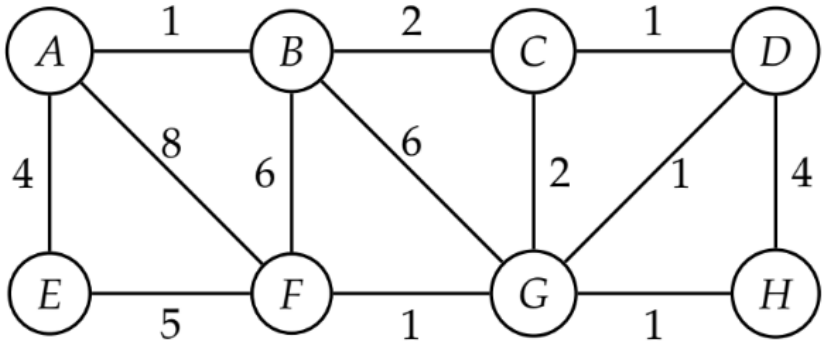
Submission Link: https://forms.gle/iVLxciz3PppPuEg78 (Deadline 11:59AM 13-10-2021)

Problem 1: Suppose Dijkstra's algorithm is run on the following graph, starting at node H. Fill in the following tables with the contents of the priority queue, which node is popped during each iteration, and the final dist(u) values. Break ties alphabetically. The first one has been done for you.



\

Iteration	Priority Queue	Node Popped
0	A(∞), B(0), C(∞), D(∞), E(∞), F(∞), G(∞), H(∞)	
1		
2		
3		
4		
5		
6		
7		
8		

u	A	B	C	D	E	F	G	H
dist(v)								

Problem 2: It is possible to have several different shortest paths between two nodes in a graph. When this is the case, the most convenient of these paths is often the one with fewest edges. For a specific starting node s, define

best(u) = the minimum number of edges in a shortest path from s to u

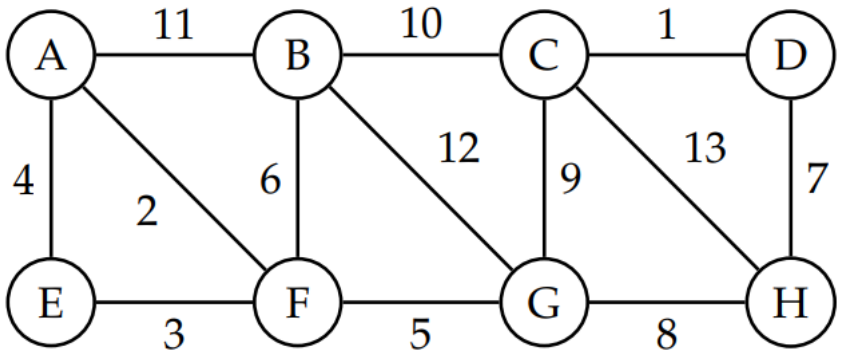
Your friend has modified Dijkstra's algorithm to compute the best(u) values. See below. Your job is to fill in the details. In particular, fill in the blanks on Lines 15 and 20.

Algorithm 1: Modified Dijkstra's Algorithm

Input: Graph $G = (V, E)$, directed or undirected; positive edge lengths $\{l_e : e \in E\}$; vertex $s \in V$.
Output: $\texttt{dist}(u)$ and $\texttt{best}(u)$, the weight and the minimum number of edges in a shortest path from s to u, respectively, for all vertices u.

```
1  Procedure dijkstra(G, l, s):
2      for all u ∈ V do
3          dist(u) ← ∞
4          best(u) ← ∞
5          prev(u) ← nil
6      end
7      dist(s) ← 0
8      best(s) ← 0
9      H ← makequeue(V) (using dist-values as keys)
10     while H is not empty do
11         u ← deletemin(H)
12         for all edges (u, v) ∈ E do
13             if dist(v) > dist(u) + l(u, v) then
14                 dist(v) ← dist(u) + l(u, v)
15                 best(v) ← _____________________
16                 prev(v) ← u
17                 decreasekey(H, v)
18             end
19             else if dist(v) = dist(u) + l(u, v) then
20                 if _____________________ then
21                     best(v) ← best(u) + 1
22                     prev(v) ← u
23                 end
24             end
25         end
26     end
```

Problem 3: Suppose Kruskal's algorithm is run on the following graph



Now fill in the following table by drawing the edges of the spanning forest in each iteration.

Iteration	Spanning Forest
1	A B C D E F G H
2	A B C D E F G H
3	A B C D E F G H
4	A B C D E F G H

Iteration	Spanning Forest
5	A B C D E F G H
6	A B C D E F G H
7	A B C D E F G H

What is the cost of the minimum spanning tree you found? _____

Draw seven edges on the following graph with six vertices, and then mark six edges with 1 and one edge with x such that an adversary can choose x such that the cost of the MST is arbitrarily large.

